

Global Function Modeling and Minimization with SOS Bayesian Optimization

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Abstract—In long-tail or novel scenarios, robots must make effective decisions with limited data. Bayesian kernel models are well-suited for this setting because they can model a wide range of cost functions from sparse samples. With a standard linear model, however, the model may be highly non-convex and difficult to minimize. We introduce sum-of-squares (SOS) kernel models, which efficiently estimate both an unknown function and its global minimum from only sparse samples. Intuitively, our approach squares the standard linear model and adds an offset corresponding to the global minimum. We show this model is compatible with a Bayesian approach and its parameters can be estimated efficiently with uncertainty. We conclude with two toy examples.

I. INTRODUCTION

In real-world settings, robots must reason in data environments where data is scarce or expensive to obtain. For example, autonomous vehicles must navigate safely through unmapped construction and other disruptions to the road [1], and disaster scenarios may include unanticipated obstacles [2]. More commonly, design problems such as hyperparameter tuning require efficient use of finite training time [3]. In this regime, Bayesian kernel models can efficiently estimate behavior on a bounded domain from sparse samples.

From a weight-space perspective, most Bayesian modeling approaches rely on a *linear kernel model*, which represents an unknown function as linear in some feature space [4]. This model is simple but powerful; for appropriate choice of feature map it can universally approximate any continuous function on a compact domain [5]. An important subset of linear kernel models are Gaussian processes (GPs), which have been used throughout robotics [6]. Despite their modeling power, reasoning over the original action space using a linear kernel model is difficult. The feature map typically introduces significant non-convexity, making it difficult to globally minimize the function model and to assess suboptimality of a candidate action.

To address this, we introduce sum-of-squares (SOS) Bayesian kernel models, which can approximate both a

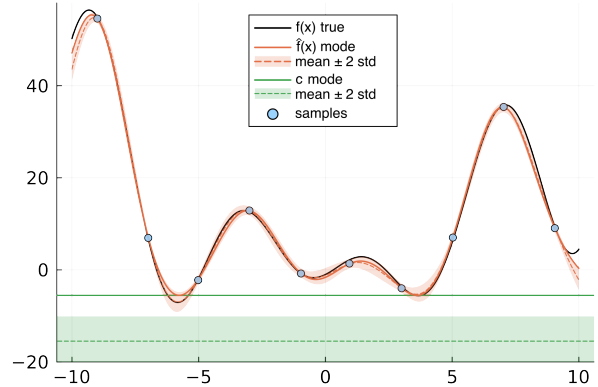


Fig. 1: A toy example of fitting a non-convex function $f(x)$ from 10 uniform samples on $[-10, 10]$. Our model gives full distributional estimates of the function and its global minimum.

function and its global minimum from a sparse set of samples. Instead of the linear model, we use the SOS model from [7], which is also universally approximating but explicitly models the global minimum. Taking a probabilistic approach, we show the posterior mode reduces to a semidefinite program (SDP) with appropriate choice of priors, and complement it with a sampling-based uncertainty estimate (Section II). The result is an expressive model of both the generating function and its global minimum which may be incorporated directly into a Bayesian optimization scheme. We demonstrate this approach in a toy example (Fig. 1) and with a simulated hopping robot in Section III.

II. SOS BAYESIAN OPTIMIZATION

Our goal is to approximate an unknown, sufficiently smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and its global minimum on a compact domain. Standard in Bayesian optimization, we split this into two parts. In the *Bayesian modeling* stage, we seek an optimal approximation of f and its minimum given noisy sample evaluations $(\mathbf{x}_i, \hat{f}(\mathbf{x}_i))$, $i = 1, \dots, N$. In the *sampling* stage, we seek a new point $\mathbf{x}_{N+1}$ to evaluate $\hat{f}(\mathbf{x}_{N+1})$ in order to obtain a better approximation of f . We focus on Bayesian modeling using sum-of-squares (SOS) tools (Section II-A) and briefly discuss sampling in Section II-B.

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A. SOS Bayesian Models

To estimate f and its minimum, we use a quadratic SOS model which exposes the global minimum:

$$f(x) \approx \hat{f}(x) = \phi(x)^\top \mathbf{A} \phi(x) + c, \quad (1)$$

where $x \in \mathbb{R}^n$, $\mathbf{A} \in \mathcal{H}$ is a positive semidefinite operator, $c \in \mathbb{R}$, and $\phi : \mathbb{R}^n \rightarrow \mathcal{H}$ is a chosen feature map. By construction, c is the global minimum of $\hat{f}$.

The modeling power of $\hat{f}$ is determined by the choice of feature map ϕ or, equivalently if we consider only inner products of features, the kernel $k(x, y) = \phi(x)^\top \phi(y)$. In particular, universal kernels such as the radial basis function can approximate arbitrary smooth functions on a compact domain [5], [7].

We find and update the parameters $\mathbf{A}, c$ with Bayesian modeling. Assuming priors on $\mathbf{A}, c$ and independent sample likelihoods, the posterior is:

$$P(\mathbf{A}, c \mid \{\tilde{f}(\mathbf{x}_i)\}_{i=1}^N) \propto P(\mathbf{A})P(c) \prod_{i=1}^N P(\tilde{f}(\mathbf{x}_i) \mid \mathbf{A}, c). \quad (2)$$

The maximizers of (2) give the maximum a posteriori (MAP) parameters for the model (1). It is difficult to maximize (2) directly because $\mathbf{A}$ may be a large or infinite-dimensional operator. Fortunately, under appropriate choice of priors, the MAP estimate $\mathbf{A}^*$ can be expressed in N dimensions using a representer theorem [7]. That is, $\mathbf{A}^* = \sum_{i=1}^N \sum_{j=1}^N \phi(\mathbf{x}_i) \mathbf{B}_{ij}^* \phi(\mathbf{x}_j)^\top$, where $\mathbf{B}^* \succeq 0$ is an $N \times N$ matrix. We pick $P(\mathbf{A})$ supported on this N -dimensional space. That is, we restrict to the kernelized function model $\hat{f}(x) = \mathbf{v}(x)^\top \mathbf{B} \mathbf{v}(x) + c$, where $\mathbf{v}(x) = [k(x, \mathbf{x}_1), \dots, k(x, \mathbf{x}_N)]^\top \in \mathbb{R}^N$.

We now specialize to a particular choice of priors and likelihood. We use a Gaussian likelihood $\tilde{f}(\mathbf{x}_i) \sim \mathcal{N}(\hat{f}(\mathbf{x}_i), \sigma_m^2)$ and an exponential prior $\gamma - c \sim \text{Exp}(\lambda)$, where $\sigma_m, \lambda \geq 0$ and γ are chosen constants. For $\mathbf{B}$, we select a Wishart prior: $\mathbf{B} \sim W_N(\mathbf{V}, N + 1)$, where $\mathbf{V} \in \mathcal{S}^N$ and $\mathbf{V} \succeq 0$. This leads to the following MAP estimator, which is a convex semidefinite program:

$$\min_{\mathbf{B} \succeq 0, c \in \mathbb{R}} \frac{1}{2} \text{tr}(\mathbf{V} \mathbf{B}) - \lambda c + \sum_{i=1}^N \frac{1}{2\sigma_m^2} z_i^2, \quad (3)$$

subject to $z_i = \tilde{f}(\mathbf{x}_i) - \mathbf{v}(\mathbf{x}_i)^\top \mathbf{B} \mathbf{v}(\mathbf{x}_i) - c$ and $c \leq \gamma$. If only the posterior mode is needed, we can safely drop the latter constraint, meaning γ does not need to be fixed a priori. This relaxation aligns (3) with [8].

The key advantage of our probabilistic derivation is that the constants and regularization are directly interpretable. Further, we can estimate the mean and variance numerically with Markov Chain Monte Carlo (MCMC) [9], [10], as shown in Fig. 1.

We emphasize that this formulation estimates both the function model *and its global minimum*. This enables

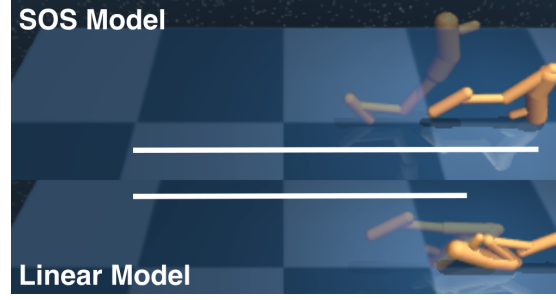


Fig. 2: Final distance of a hopper optimized by SOS Bayesian optimization and standard Bayesian optimization.

certifiable minimization of the surrogate and a full posterior distribution over the function minimum, despite any non-convexity. We may also extract the approximate minimizer by analogy to SOS [8], [11].

B. Sampling Approaches

Our approach is compatible with standard Bayesian optimization sampling techniques [12]. Of particular interest is Thompson sampling [13], where the next point is the minimizer of a random draw from the model posterior (*e.g.*, from MCMC). Since SOS kernel models explicitly model the global minimum, we can directly certify that a candidate minimizer is globally optimal.

III. EXPERIMENTS

We validate our SOS kernel model in a 1D toy example (Fig. 1) and to optimize a simulated robot's distance (Fig. 2). In the 1D toy example, we model the non-convex and non-polynomial function $f(x) = x^2/3 - 3x \sin(x - 2) - \cos(x)$ over the domain $x \in [-10, 10]$ and take 10 equidistant samples with Gaussian noise. Kernel parameters were tuned to slightly exceed the sample width. As Fig. 1 shows, this gives an accurate function model, including an estimate of its minimum.

We also use SOS Bayesian optimization to maximize the displacement of a Mujoco [14] hopping robot. We fix a one-parameter controller $f_i(x) = \sin(2\pi tx + \pi i/2)$, $i = 0, \dots, 3$ for each joint. Fig. 2 shows the optimized policy using an SOS model and a linear model with the same parameters and Thompson sampling. The SOS model finds a crawling policy which performs slightly better than the somersault found by the linear model.

IV. CONCLUSION

We presented SOS Bayesian kernel models, which enable estimating an unknown function and its global minimum with uncertainty from a sparse set of samples. Future work will evaluate the SOS model beyond 1D toy examples and rigorously compare its performance with the linear model, including the suboptimality of the final estimate. Important intermediate steps include replacing the brittle MCMC for specific choice of priors and tuning kernel parameters using cross-validation.

REFERENCES

- [1] M. Hallgarten, J. Zapata, M. Stoll, K. Renz, and A. Zell, "Can vehicle motion planning generalize to realistic long-tail scenarios?" In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2024, pp. 5388–5395.
- [2] K. Nagatani et al., "Emergency response to the nuclear accident at the Fukushima daiichi nuclear power plants using mobile rescue robots," *Journal of Field Robotics*, vol. 30, no. 1, pp. 44–63, 2013.
- [3] J. Snoek, H. Larochelle, and R. P. Adams, "Practical bayesian optimization of machine learning algorithms," *Advances in neural information processing systems*, vol. 25, 2012.
- [4] C. Rasmussen and C. Williams, *Gaussian Processes for Machine Learning*. The MIT press, Cambridge, MA, 2006.
- [5] C. A. Micchelli, Y. Xu, and H. Zhang, "Universal kernels.," *Journal of Machine Learning Research*, vol. 7, no. 12, 2006.
- [6] M. P. Deisenroth, D. Fox, and C. E. Rasmussen, "Gaussian processes for data-efficient learning in robotics and control," *IEEE transactions on pattern analysis and machine intelligence*, vol. 37, no. 2, pp. 408–423, 2013.
- [7] U. Marteau-Ferey, F. Bach, and A. Rudi, "Non-parametric models for non-negative functions," *Advances in neural information processing systems*, vol. 33, pp. 12 816–12 826, 2020.
- [8] A. Rudi, U. Marteau-Ferey, and F. Bach, "Finding global minima via kernel approximations," *arXiv preprint arXiv:2012.11978*, 2020.
- [9] M. D. Hoffman, A. Gelman, et al., "The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1593–1623, 2014.
- [10] T. E. Fjelde et al., "Turing.jl: A general-purpose probabilistic programming language," *ACM Trans. Probab. Mach. Learn.*, Feb. 2025, Just Accepted. DOI: 10.1145/3711897.
- [11] A. Groudiev, F. Schramm, É. Berthier, J. Carpentier, and F. Dümbgen, "Sampling-based global optimal control and estimation via semidefinite programming," *arXiv preprint arXiv:2507.17572*, 2025.
- [12] R. Garnett, *Bayesian Optimization*. Cambridge University Press, 2023.
- [13] W. R. Thompson, "On the likelihood that one unknown probability exceeds another in view of the evidence of two samples," *Biometrika*, vol. 25, no. 3/4, pp. 285–294, 1933.
- [14] E. Todorov, T. Erez, and Y. Tassa, "Mujoco: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, IEEE, 2012, pp. 5026–5033. DOI: 10.1109/IROS.2012.6386109.